RAUFASTE Christophe
Professor
Institut de Physique de Nice (INPHYNI)
Université Côte d'Azur
17 rue Julien LAUPRÊTRE, 06200 Nice

Born on September 20th, 1980 in Hayange
French, 1 child
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web: http://sites.unice.fr/site/raufaste/raufaste/

EDUCATION

2017: "Habilitation à diriger les recherches", Université Côte d'Azur.
"Dynamique d'objets capillaires et écoulement aux interfaces libres".
committee: I. Cantat, C. Clanet, N. Vandewalle, F. Graner, M. Argentina, E. Villermaux
2004-2007: PhD supervised by F. Graner, Université Grenoble I.
"Rhéologie et imagerie des écoulements 2D de mousse : approche expérimentale, numérique et théorique".
2003-2004: Master's degree in physics of liquids, Paris
2002-2003: "Agrégation de physique"
2000-2004: Student at ENS Lyon, physics department

PROFESSIONAL EXPERIENCE

since 2022: Full Professor, Université Côte d'Azur
2009-2022: Associate Professor, Université Nice Sophia Antipolis, section CNU 28.
2018: sabbatical, 8 months, DAMTP, Cambridge, group of Pr. Ray Goldstein
CNRS sabbatical, 6 months in 2014/2015 and 2015/2016
2007-2009: postdoc, group "Physics of Geological Processes", Univ. Oslo
2004-2007: PhD and teaching assistant, Université Grenoble I

AWARDS AND HONORS

2020-2025: junior fellow of the Institut Universitaire de France
2023: Milton van Dyke award shared with collaborators for the video "Dynamics of frost propagation" (gallery of fluid motion 2023)
2021: Research excellence award, Université Côte d'Azur
2016-2020: PEDS (bonus for doctoral supervision)
2012-2016: PES (bonus for scientific excellence)
2008: PhD prize awarded by the French Rheology Society

SUPERVISION

Postdocs: Florence Haudin (2010-2011), Florence Gibouin (2016-2017), Li Fu (2020-2022), Ludovic Keiser (2022-2023), Guillaume Giombini (2023), Antoine Monier (2022-2024)
PhD students: Geoffroy Kirstetter (2010-2014), Alexandre Cohen (2011-2015), Christophe d'Angelo (2018-2021), Jesus Sanchez (2018-2021), Guillaume Giombini (2019-2023), Florian Schott (2020-2025, Lunds Univ.), David Paulovics (2021-2025), Alexandre Vigna–Brümmer (2023-)
Master students: Mathieu Pellegrin (M2, 2011), Damien Scandola (M1, 2013), Marine Borocco (M2, 2014), Alexandre Barzyk (M1, 2014), Abidi Bilel (M1, 2016), Christophe d'Angelo (M1, 2017), Christophe d'Angelo (M2, 2018), Guillaume Giombini (M2, 2019), William Bontemps (M1, 2019), Lisa

Sicard (M2, 2020), Simeon Djambov (M1, 2020), Leo Guschemann (M1, 2020), David Paulovics (M2, 2021), François-Xavier Gauci (M2, 2022), Alexandre Vigna-Brummer (M2, 2023), Mathieu Londiche (M1, 2024), Karen Trevia (M2, 2025)

ADMINISTRATION

2024-2028: elected member of the COSP (COnseil Scientifique et Pédagogique), EUR SPECTRUM
2021-2023: deputy director of the Physics Institute of Nice (INPHYNI)
since 2021: member of the steering committee of the GDR MePhy
https://mephysociety.wordpress.com/
since 2020: team leader of the "Complex Fluids" group
since 2020: member of the scientific council of the Physics Institute of Nice
since 2018: member of the steering committee of the Master OAM ("Ondes, Atomes et Matière", Université Côte d'Azur). http://master-oam.unice.fr/
2015-2021: elected member of the standing committee on human resources in physics (Univ. Nice): advisory body that advises on the recruitment strategy (Associate Professors and Professors) in physics at the University of Nice.
2011-2020: responsible for the PhD-students at the institute and at the doctoral school.
2011-2020: elected member of the Institute Council

RESEARCH MANAGEMENT

since 2020: team leader of the "Complex Fluids" group (7 permanent researchers, 2 postdocs, 3 PhD students in average)
since 2017: principal investigator (with F. Celestini) of the superpropulsion research program. 3 senior researchers, 1 researcher from MSC and 1 theoretical physicist from the Niels Bohr Institute involved, 2 PhDs funded since 2018, 8 articles published
since 2011: principal investigator of the liquid foam program. 5 senior researchers involved, collaborations with researchers from ENS Lyon, LIPHY (Grenoble), LPS (Orsay) and IPR (Rennes) as well as with researchers from Cambridge, Lund and Aberystwyth Universities and from the Paul Scherrer Institute, 2 PhDs defended, 2 PhD in progress, 1 postdoc, 20 articles published, 1 keynote and 1 invited presentation in international conference, 1 invited lecture

PROJECTS FUNDED AS PRINCIPAL INVESTIGATOR

2023-2026: "Etude d'effets de flottabilité 2D dans un film de savon". 1 PhD funded by the Ministry of Higher Education and Research (MESR), 120 k€.
2022-2024: "Dynamique d'interfaces libres en mécanique des fluides avec une application en physique des plantes et une en micro fluidique fondamentale et appliquée". 1 postdoc funded by the Idex UCA Jedi by the call AIDES INDIVIDUELLES JEUNES CHERCHEURS 2021, 107 k€.
2020-2024: "Stability of foam films: marginal regeneration and drainage (DrainFilm)" funded by the ANR, three partners (INPHYNI, LPS Orsay, IPR Rennes), 411k€.
2019-2022: "Use of the superpropulsion effect to optimize ballistic throws". 1 PhD funded by the Directorate General of Armaments, 118k€.
2018-2020: "Fast infrared imaging for thermohydrodynamics". 95k€ obtained from several sources (CNRS/UNS/Doeblin/RégionPACA) to get an infrared high-speed camera.

2010-2016: 3 projects about X-ray tomography experiments on foam flow. French-Swiss collaboration, four partners (INPHYNI, SLS, ENS LYON, LIPHY), 5 researchers.

"Resolving three dimensional foam flow", 2015-2016, 49k€ in beam time.

"Structural rearrangements and flow in 3D liquid foams", 2011-2013, 59k€ in beam time.

"Dynamics of gas injection inside 3D liquid foams", 2010-2011, 49k€ in beam time.

PARTNER OF FUNDED PROJECTS

2021-2025: "Impinging micro-jets", PI M. Argentina (UCA). 418 k€ funded by the ANR, two partners (INPHYNI, IEMN), 6 researchers.

2020-2025: "Multi-scale analysis of structural rearrangements in a flowing liquid foam using X-ray tomographic microscopy", PI R. Mokso (MaxIV, Lund University). 355 k€ from the Swedish Research Council (SRC). X-ray tomography experiments and analysis on foam flow. French-Swedish collaboration, four partners (MaxIV, INPHYNI, ENS LYON, LIPHY), 6 researchers.

2020-2023: "Model-based and deep learning methods for robotic control", PI G. Allibert. 190k€ (UCA IDEX). Transdisciplinary project between the physics and informatics institutes of Nice. Two partners (INPHYNI, I3S), 3 researchers. Responsible for the experiments with the robotic fish.

2016-2018: "Swimming Underwater roBOT", PI M. Argentina. 77k€ (UCA IDEX), 3 researchers. Responsible for the experiments.

2010-2013: "Cavitation and bubble dynamics", PI X. Noblin. 223k€ (ANR), 4 researchers. Responsible for 1 part of 3.

MOBILITY AND VISITS FOR COLLABORATION

2024: Institut Charles Sadron, Strasbourg, collaboration W. Drenckhan (1 month)
2019: Lavrentyev Institute of Hydrodynamics, Russia, collaboration E. Ermanyuk (2 weeks)
2018: DAMTP, University of Cambridge, sabbatical, group of Pr. R. Goldstein (8 months)
2018: Aberystwyth University, UK, collaboration S. J. Cox (2 weeks)
2012-2016: Niels Bohr Institute, collaboration J. Mathiesen (2x 1 week)
2011-2019: ENS Lyon, collaboration S. Santucci (4x 1 week)
2011-2016: IPR Rennes, collaboration B. Dollet (3x1 week)
Sep. 2011: PSI Villigen, Switzerland, collaboration R. Mokso (2 weeks)
2008-2009: University of Oslo, Norway, postdoc (2 years)
Jun. 2007: Aberystwyth University, UK, collaboration S. J. Cox (2 weeks)
Mar. 2005: Los Alamos National Laboratory, US, collaboration Yi Jiang (1 month)

PHD COMMITTEES (19 external committees, 12 as reviewer)

2024 (examiner) Nicolo Galvani, NAVIER - Université Gustave Eiffel
2024 (reviewer) Friedrich Walzel, ICS - Université de Strasbourg
2024 (examiner) Sardor Israilov, INPHYNI - Université Côte d'Azur
2023 (examiner) Rodolphe Grivet, LadHyX, Institut Polytechnique de Paris
2023 (reviewer) Jouanlanne Manon, ICS - Université de Strasbourg
2023 (examiner) Metzmacher Jean, GRASP - Université de Liège
2022 (reviewer) Yutong Cui, ESPCI - Sorbonne Université
2022 (president) Maxime Raboisson-Michel, INPHYNI - Université Côte d'Azur
2022 (examiner) Lauren Rose, LP ENS Lyon - ENS Lyon

2022 (examiner) Chiara Guidolin, LPS Orsay - Université Paris Saclay
2022 (reviewer) Aditya Jha, PMMH - PSL - ESPCI
2021 (reviewer) Carlos Arauz Moreno, LIPHY Grenoble - Université Grenoble-Alpes
2020 (reviewer) Manon Marchand, LPS Orsay - Université Paris Saclay
2020 (reviewer) Lucile Favreau, LP ENS Lyon - ENS Lyon
2019 (reviewer) Jonas Miguet, LPS Orsay - Université Paris Saclay
2019 (reviewer) Quentin Raimbaud, IPR Rennes - Université de Rennes
2019 (examiner) Goce Koleski, LOMA - Université de Bordeaux
2019 (examiner) David Gross, INPHYNI - Université Côte d'Azur
2018 (reviewer) Armelle Keiser, PMMH - PSL - ESPCI
2018 (reviewer) Quentin Ehlinger, ILM - Université de Lyon
2014 (examiner) Sébastien Kosgodagan Acharige, MSC - Université Paris Diderot
2013 (reviewer) Kevin Mader, ETH Zürich

HDR COMMITTEES
2023 (reviewer) Caroline Derec, MSC, Université Paris Cité

RECRUITMENT COMMITTEES (2 external and 4 local committees))
Professeur des universités, article 46-4, Université Côte d'Azur, sections 29/30,
recrutement 01/09/2024, référence GALAXIE 250
Maître de conférences, Ecole Centrale Méditerranée, sections 28/29/30,
recrutement 01/09/2023, référence GALAXIE 4036
Professeur des universités, article 46-3, Université Gustave Eiffel, section 28,
recrutement 01/09/2023, référence GALAXIE 96
Professeur des universités, article 46-4, Université Côte d'Azur, section 63,
recrutement 01/09/2023, référence GALAXIE 197
Maître de conférences, article 26-I-1, Université Côte d'Azur, section 28/29,
recrutement 01/09/2020, référence GALAXIE 14.
Maître de conférences, article 26-1, Université Côte d'Azur, section 28,
recrutement 01/09/2017, N°emploi ID451.

ORGANIZATION OF CONFERENCES
MephyDay "Environmental mechanics", 12 June 2024, Paris
https://mephysociety.wordpress.com/mephy-day-environmental-mechanics/
Member of the organization and scientific committee
Rencontres du non linéaire, 18-20 March 2024, Paris
http://nonlineaire.univ-lille1.fr/SNL/
Member of the scientific committee
MephyDay "surface tension, geometry and soft solids", 11 Oct. 2023, Paris
https://mephysociety.wordpress.com/mephy-day-surface-tension-geometry-and-soft-solids/
Member of the organization and scientific committee
Rencontres du non linéaire, 28-30 March 2023, Paris
http://nonlineaire.univ-lille1.fr/SNL/
Member of the scientific committee

Rencontres du non linéaire, 29-31 March 2022, Paris
http://nonlineaire.univ-lille1.fr/SNL/
Member of the scientific committee
Fluids and Complexity II, 29-30 Nov. 2021, Nice
https://fcomplexity2.sciencesconf.org/
Member of the scientific committee
Waves Côte d'Azur, 4-7 June 2019, Nice
http://wavescotedazur.org/
Member of the organization and scientific committees
Workshop "Mousses, films et effets Marangoni", 27-29 March 2019, Rennes
https://ipr. univ-rennes1.fr/agenda/workshop-mousses-films-et-effets-marangoni
Member of the scientific committee

LECTURES, COMMUNICATIONS, ABSTRACTS, SEMINARS
2 invited lectures (summer school and University)
1 keynote and 1 invited presentation at international conferences
1 invited presentation at a national conference
18 selected oral presentations at international conferences
16 selected oral presentations at national conferences
120 abstracts at national and international conferences
20 invited seminars in institutes (11 in France and 9 abroad)

PUBLICATIONS

Orcid ID : 0000-0003-4328-7438
ResearcherID : R-2217-2017
57 articles published

INTERNATIONAL PEER-REVIEWED JOURNALS
[57] Friction-induced bubble edge curvature in flowing two-dimensional confined foams
C. Raufaste, L. Rose, S. Santucci and B. Dollet,
Phys. Rev. Fluids 10, 023301 (2025).

[56] Throwing of slender elastic projectiles
G. Giombini, F. Celestini and C. Raufaste,
Phys. Rev. E 111, 015504 (2025).
The paper is featured as a Research Highlight in Nature magazine: How a wobbly arrow can achieve superpropulsion.

[55] Self-similar and universal dynamics in drainage of mobile soap films A. Monier, F.-X. Gauci, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste, Phys. Rev. Fluids 9, 124001 (2024).
[54] High-speed and infrared imaging of frost propagation on breath figures
D. Paulovics, C. Raufaste, T. Frisch, C. Claudet and F. Celestini,
Phys. Rev. Fluids 9, 110507 (2024).

This paper is associated with a video winner of a 2023 American Physical Society's Division of Fluid Dynamics (DFD) Milton van Dyke Award for work presented at the DFD Gallery of Fluid Motion. The original video is available online at the Gallery of Fluid Motion, https://doi.org/10.1103/APS.DFD.2023.GFM.V0079.

53] Leidenfrost jets
D. Paulovics, T. Frisch, C. Raufaste and F. Celestini,
Phys. Rev. Fluids 9, L112001 (2024).
The paper is listed as one of the editors' suggestions. It is featured in NewScientist: Jets of liquid bounce off hot surfaces without ever touching them.

[52] Quantifying order in breath figure patterns through Voronoi entropy D. Paulovics, E. Bormashenko, C. Raufaste and F. Celestini, Phys. Rev. E 110, 024302 (2024).

[51] Scaling the tail beat frequency and swimming speed in underwater undulatory swimming
J. Sanchez-Rodriguez, C. Raufaste and M. Argentina,
Nat Commun 14, 5569 (2023).

[50] Impact of composite soft-rigid elastic projectiles: A numerical study
C. D'Angelo, G. Giombini, F. Celestini and C. Raufaste,
Phys. Rev. E 108, 015001 (2023).

[49] Three-dimensional liquid foam flow through a hopper resolved by fast X-ray microtomography
F. Schott, B. Dollet, S. Santucci, C. Claudet, M. Argentina, C. Raufaste and R. Mokso
Soft Matter 19, 1300--1311 (2023).
The paper is selected for the Soft Matter cover, volume 19, issue 7, Feb. 2023

[48] Reinforcement learning approach to control an inverted pendulum: A general framework for educational purposes
S. Israilov, L. Fu, J. Sánchez-Rodríguez, F. Fusco, G. Allibert, C. Raufaste and M. Argentina
PLOS ONE 18(2), e0280071 (2023).

[47] Atomistic Description of Interdroplet Ice-Bridge Formation during Condensation Frosting
S. Curiotto, D. Paulovics, C. Raufaste, F. Celestini, T. Frisch, F. Leroy, F. Cheynis and P. Müller,
Langmuir 39, 579--587 (2023).

[46] Redistribution of translational kinetic energy: the surprises of bounces
C. D'Angelo, L. Viennot, M. Argentina, F. Celestini and C. Raufaste,
Eur. J. Phys. 43, 065702 (2022).

[45] Collapse of a hemicatenoid bounded by a solid wall: instability and dynamics driven by surface Plateau border friction
C. Raufaste, S. Cox, R. E. Goldstein and A. I. Pesci,
Soft Matter 18, 4944--4952 (2022).

[44] Dynamics of Frost Propagation on Breath Figures
D. Paulovics, C. Raufaste, T. Frisch, C. Claudet and F. Celestini,
Langmuir 38, 2972–2978 (2022).

[43] Use of compliant actuators for throwing rigid projectiles
G. Giombini, J.Mathiesen, C. D'Angelo, M. Argentina, C. Raufaste and F. Celestini,
Phys. Rev. E 105, 025001 (2022).

[42] Geometrical Evolution of Surfaces Dynamics of the Gaussian Curvature of a Collapsing Soap Film
R. E. Goldstein, A. I. Pesci, C. Raufaste and J. D. Shemilt,
Phys. Rev. E 104, 035105 (2021).

[41] Actuating water droplets on liquid infused surfaces: A rickshaw for droplets
C. Raufaste, S. J. Cox and F. Celestini,
Phys. Rev. Fluids. 6, 083603 (2021).

[40] Effect of gravity on the orientation and detachment of cubic particles adsorbed at soap film or liquid interfaces
I. T. Davies and C. Raufaste,
Soft Matter 17, 6964--6971 (2021).

[39] Proprioceptive mechanism for bioinspired fish swimming
J. Sanchez-Rodriguez, F. Celestini, C. Raufaste and M. Argentina,
Phys. Rev. Lett. 126, 234501 (2021).
The paper is listed as one of the editors' suggestions. It is featured in Physics: Robo-Fish Replicates Real Swimming Action and in Phys.org: A proprioceptive mechanism to enable fish-like swimming in robots.

[38] Impact dynamics of composite elastorigid projectiles onto solid surfaces
C. D'Angelo, L. Viennot, M. Argentina, F. Celestini and C. Raufaste,
Phys. Rev. E 103, 053005 (2021).

[37] Drag analysis with a self-propelled flexible swimmer
D. Gross, Y. Roux, C. Raufaste and M. Argentina,
Phys. Rev. Fluids 6, 053101 (2021).

[36] Contact layer as a propelling advantage in throwing
F. Celestini, J. Mathiesen, M. Argentina and C. Raufaste,
Phys. Rev. Applied 14, 044026 (2020).
The paper is featured in Physics: Springy Material Boosts Projectile Performance and in Le Monde: Du gel pour transformer les solides en excellents sauteurs.

[35] Quasi two-dimensional foam flow through and around a permeable obstacle

N. Shmakova, T. Chevalier, A. Puisto, M. Alava, C. Raufaste and S. Santucci,
Phys. Rev. Fluids 5, 093301 (2020).
The paper has led to a book chapter in: Médici, E.F., Otero, A.D. (eds) Album of Porous Media. Springer, Cham (2023).

[34] A minimal model of self propelled locomotion
J. Sanchez-Rodriguez, C. Raufaste and M. Argentina,
J. Fluids Struct. 125, 103071 (2020).

[33] Stress-Induced Dinoflagellate Bioluminescence at the Single Cell Level
M. Jalaal, N. Schramma, A. Dode, H. de Maleprade, C. Raufaste, R. E. Goldstein,
Phys. Rev. Lett. 125, 028102 (2020).
The paper is listed as one of the editors' suggestions. It is featured in Physics: Flashes bright when squeezed tight: How single-celled organisms light up the oceans and in Physics Today: Mechanically stressed phytoplankton light up.

[32] Ferrofluid Leidenfrost droplets.
C. D'Angelo, C. Raufaste, P. Kuzhir, F. Celestini.
Soft Matter 15, 5945–5950 (2019).

[31] Experimental characterization of droplet adhesion : the Ejection Test Method (ETM) applied to surfaces with various hydrophobicity.
G. Ramos Chagas, F. Celestini, C. Raufaste, A. Gaucher, D. Prim, S. Amigoni, F. Guittard, T. Darmanin.
J. Phys. Chem. A 122, 8693–8700 (2018).

[30] Study of the thrust-drag balance with a swimming robotic fish.
F. Gibouin, C. Raufaste, Y. Bouret, M. Argentina.
Phys. fluids 30, 091901 (2018).

[29] Foam flows through a local constriction.
T Chevalier, J. Koivisto, N. Shmakova, M. J. Alava, A Puisto., C. Raufaste, S. Santucci.
J. Phys. : Conf. Ser. 925, 012025 (2017).

[28] Melde's experiment on a vibrating liquid foam microchannel.
A. Cohen, N. Fraysse, C. Raufaste.
Phys. Rev. Lett. 119, 238001 (2017).
The paper is featured in Physics with a Focus story written by Michael Schirber: Liquid String Vibrations.

[27] Superpropulsion of droplets and soft elastic solids.
C. Raufaste, G. Ramos Chagas, T. Darmanin, C. Claudet, F. Guittard, F. Celestini.
Phys. Rev. Lett. 119, 108001 (2017).

The paper is listed as one of the editors' suggestions from Physical Review Letters and is featured in Physics with a Focus story written by Philip Ball: Superpropulsion of liquid drops. Other highlights in Physics World (IOP), Pour la Science and Science & vie.

[26] Solitary-like waves in a liquid foam microchannel.
Y. Bouret, A. Cohen, N. Fraysse, M. Argentina and C. Raufaste. Phys. Rev. F 1, 043902 (2016).

[25] Bubble dynamics inside an outgassing hydrogel confined in a Hele-Shaw cell.
F. Haudin, X. Noblin, Y. Bouret, M. Argentina and C. Raufaste.
Phys. Rev. E 94, 023109 (2016).

[24] Reactive Leidenfrost droplets.
C. Raufaste, Y. Bouret and F. Celestini.
Europhys. Lett. 114, 46005 (2016).
The paper is listed as one of the editors' choices for 2016

[23] Scale selection in columnar jointing.
A. Christensen, C. Raufaste, M. Misztal, F. Celestini, M. Guidi, C. Ellegaard and J. Mathiesen.
J. Geophys. Res. Solid Earth 121, 1462–1482 (2016).

[22] Three-dimensional foam flow resolved by fast X-ray tomographic microscopy.
C. Raufaste, B. Dollet, K. Mader, S. Santucci and R. Mokso.
Europhys. Lett. 111, 38004 (2015).

[21] Drop coalescence and liquid flow in a single Plateau border.
A. Cohen, N. Fraysse and C. Raufaste.
Phys. Rev. E 91, 053008 (2015).

[20] Hole growth dynamics in a two dimensional Leidenfrost droplet.
C. Raufaste, F. Celestini, A. Barzyk and T. Frisch.
Phys. fluids 27, 031704 (2015).

[19] One dimensional capillary jumps.
M. Argentina, A. Cohen, Y. Bouret, N. Fraysse and C. Raufaste.
J. Fluid Mech. 765, 1–16 (2015).
The paper was selected for the J. Fluid Mech. cover, issue 765, Feb. 2015

[18] Rheology of aqueous foams. B. Dollet and C. Raufaste. C. R. Phys. 15, 731–747 (2014).

[17] Inertial mass transport and capillary hydraulic jump in a liquid foam microchannel.
A. Cohen, N. Fraysse, J. Rajchenbach, M. Argentina, Y. Bouret and C. Raufaste.
Phys. Rev. Lett. 112, 218303 (2014).
The paper is listed as one of the editors' suggestions from Physical Review Letters and is highlighted by the Physics Institute of the CNRS.

[16] Two dimensional Leidenfrost droplets in a Hele Shaw cell.
F. Celestini, T. Frisch, A. Cohen, C. Raufaste, L. Duchemin and Y. Pomeau.
Phys. fluids 26, 032103 (2014).
The paper is listed as one of the research highlights from Physics of Fluids

[15] Deformation of a free interface pierced by a tilted cylinder : variation of the contact angle.
C. Raufaste and S. Cox.
Colloids Surf. A 438, 126–131 (2013).

[14] Quantitative 3D characterization of cellular materials : segmentation and morphology of Foam.
K. Mader, R. Mokso, C. Raufaste, B. Dollet, S. Santucci, J. Lambert, M. Stampanoni.
Colloids Surf. A 415, 230–238 (2012).

[13] Deformation of a free interface pierced by a tilted cylinder.
C. Raufaste, G. Kirstetter, F. Celestini and S. J. Cox.
Europhys. Lett. 99, 24001 (2012).

[12] Jet impact on a soap film. G. Kirstetter, C. Raufaste and F. Celestini.
Phys. Rev. E 86, 036303 (2012).
The study inspired the problem Jet and film in the Young Physicists Tournament 2013

[11] The mechanism of porosity formation during solvent-mediated phase transformations.
C. Raufaste, B. Jamtveit,T. John, P. Meakin and D. K. Dysthe.
Proc. R. Soc. London A 467, 1408–1426 (2011).

[10] Understanding and predicting viscous, elastic, plastic flows.
I. Cheddadi, P. Saramito, B. Dollet, C. Raufaste and F. Graner.
Eur. Phys. J. E 34, 1–15 (2011).

[9] Morphological transitions in partially gas-fluidized granular mixtures.
A. Nermoen, C. Raufaste, S. D. de Villiers, E. Jettestuen, P. Meakin, D. K. Dysthe.
Phys. Rev. E 81, 061305 (2010).

[8] Discrete rearranging disordered patterns: Prediction of elastic and plastic behavior, and application to two-dimensional foams.
C. Raufaste, S. J. Cox, P. Marmottant and F. Graner.
Phys. Rev. E 81, 031404 (2010).

[7] Dissipation in quasi-two-dimensional flowing foams.
C. Raufaste, A. Foulon and B. Dollet.
Phys. fluids 21, 053102 (2009).

[6] Numerical modelling of foam Couette flows.

I. Cheddadi, P. Saramito,C. Raufaste, P. Marmottant and F. Graner.
Eur. Phys. J. E 27, 123–133 (2008).

[5] Discrete rearranging disordered patterns, part II: 2D plasticity, elasticity and flow of a foam.
P. Marmottant, C. Raufaste and F. Graner.
Eur. Phys. J. E 25, 371–384 (2008).

[4] Discrete rearranging disordered patterns, part I: Robust statistical tools in two or three dimensions.
F. Graner, B. Dollet, C. Raufaste and P. Marmottant.
Eur. Phys. J. E 25, 349–369 (2008).

[3] Yield drag in a 2D foam flow around a circular obstacle: Effect of liquid fraction.
C. Raufaste, B. Dollet, S. J. Cox, Y. Jiang and F. Graner.
Eur. Phys. J. E 23, 217-228 (2007).

[2] 2D flow of foam around obstacles: force measurements.
B. Dollet, F. Elias, C. Quilliet, C. Raufaste, M. Aubouy and F. Graner.
Phys. Rev. E 71, 031403 (2005).

[1] Penetration of negatively buoyant jet in a miscible liquid.
P. Philippe, C. Raufaste, P. Kurowski and P. Petitjeans.
Phys. fluids 17, 053601 (2005).

INVITED ARTICLE IN A NATIONAL JOURNAL
Physique du lancer
F. Celestini and C. Raufaste
To appear in Reflets de la physique 2025

Rhéologie et imagerie des écoulements 2D de mousse.
C. Raufaste.
Rhéologie 14, 13-14 (2008).

INVITED REVIEW
Rheology of aqueous foams.
B. Dollet and C. Raufaste.
C. R. Phys. 15, 731–747 (2014).

THESES
PhD
Rhéologie et imagerie des écoulements 2D de mousse : approche expérimentale, numérique et théorique. Université de Grenoble, 2007.

Committee: S.J. Cox, I. Cantat, S. Cohen-Addad, A. Ammar, F. Graner.
http://tel.archives-ouvertes.fr/tel-00193248/fr/

“Habilitation à diriger les recherches”

Dynamique d’objets capillaires et écoulement aux interfaces libres. Univ. Côte d’Azur, 2017.

committee: I. Cantat, C. Clanet, N. Vandewalle, F. Graner, M. Argentina, E. Villermaux

https://tel.archives-ouvertes.fr/tel-01657142

LECTURES, PRESENTATIONS, SEMINARS, ABSTRACTS

INVITED LECTURES

April 2021, Films, Bubbles and Foam, University of Liège, one week invited lecture
https://www.cesam.uliege.be/cms/c_7055854/fr/lecon-inaugurale-19/04/21-films-bubbles-and-foam-par-ass-prof-christophe-raufaste-universite-cote-d-azur-nice
June 2014, Gouttes en mouillage nul, summer school “aux rencontres de Peyresq”.

KEYNOTE AND INVITED PRESENTATIONS AT INTERNATIONAL CONFERENCES

Jul. 2018 (keynote) EUFOAM 2018, Liège, Belgium. Stability and relaxation dynamics of a single Plateau border.
Sep. 2011 (invited) Jump 2011, PSI user meeting, Villigen, Switzerland. Imaging and mechanics of liquid foam flows.

INVITED PRESENTATION AT A NATIONAL CONFERENCE

Jul. 2023, Congrès annuel de la Société Française de Physique. Throwing soft objects with the superpropulsion effect.

INVITED SEMINARS

Feb. 2024, IUSTI, Marseille
Jul. 2023, Institut Jean Le Rond d'Alembert, Paris
May 2023, Institut Charles Sadron, Strasbourg
Dec. 2020, Institut de Physique de Nice
Jul. 2019, Lavrentyev Institute of Hydrodynamics, Novosibirsk
Apr. 2019, LiPhy, Grenoble
Nov. 2018, Laboratoire Matière et Systèmes Complexes, Paris
Feb. 2018, University of Cambridge, DAMTP
Feb. 2018, University of Aberystwyth
Jan. 2017, Observatoire de la Côte d'Azur, Nice
May 2016, Institut de Physique de Rennes
Apr. 2016, ENS Lyon
Apr. 2016, Niels Bohr Institute, Copenhagen
Nov. 2014, Matière et Systèmes Complexes, Paris
Dec. 2012, Niels Bohr Institute, Copenhagen
Jan. 2011, Institut de Physique de Rennes
Feb. 2010, Laboratoire J.A. Dieudonné, Nice
Feb. 2010, Laboratoire de Physique de la Matière Condensée, Nice
Jun. 2007, Physics of Geological Processes, Oslo
May 2007, Institute of Chemistry, Graz

SELECTED PRESENTATIONS AT INTERNATIONAL CONFERENCES

Oct. 2024, INES 2024, Nice, Shape and dynamics of menisci at the contact with soap films
Sep. 2024, EFDC1 2024, Aachen, Drainage of mobile soap films under gravity: characterizing the space-time dynamics

July 2024, Eufoam 2024, Dresden, Space-time dynamics in gravity drainage of mobile soap films
July 2022, Eufoam 2022, Krakow, Coupling the shape and motion of bubbles in a confined 2D flowing foam.
Jul. 2020, Eufoam 2020, Aberystwyth (virtual), Geometrical evolution during the collapse of soap-film catenoids.
Jun. 2019, Waves Côte d'Azur, Nice, Nonlinear waves in Plateau borders.
Jul. 2018, Eufoam 2018, Liège, Stability and relaxation dynamics of a single Plateau border.
Nov. 2017, APS DFD Denver, Hydraulic jumps in the liquid foam microchannels.
Mar. 2015, FloMat, Rome, Inertial flows in liquid foam microchannels.
Jul. 2012, Eufoam 2012, Lisbon, Jet refraction across a soap film.
Jan. 2012, CECAM workshop, dissipative rheology of foams, Dublin, Drop coalescence along a Plateau Border.
Jan. 2012, CECAM workshop, dissipative rheology of foams, Dublin, Impact of a liquid jet on a soap film.
Sep. 2011, Jump 2011, Villigen, Imaging and mechanics of liquid foam flows.
Sep. 2011, Liquid Matter, Vienna, Interaction of a liquid jet with a soap film.
May 2011, IUTAM Symposium : Mechanics of liquid and solid foams, Austin, US. Interaction of a liquid jet with a liquid foam: a film scale study.
Jul. 2010, Eufoam, Borovets, Bulgaria, Orientational effects in the flow of liquid foams.
Jun. 2010, Cryspom II, Brienz, Switzerland, Crystal growth and porosity formation during solvent-mediated phase transformations.
Jan. 2010, 29th Nordic Geological Winter Meeting, Oslo, Mechanisms of permeability enhancement during chemical alteration.

SELECTED PRESENTATIONS AT NATIONAL CONFERENCES
June 2024, Soft Matter meeting 2024, Strasbourg
Jul. 2023, Congrès annuel de la SFP, Paris
June 2023, Workshop Mousses, films et effets Marangoni, Rennes
Mar 2022, Workshop systèmes à 3 phases du GDR MEPHY, Paris
Dec. 2018, Fluids and complexity, Nice
Dec. 2017, Nice Nonlinearities, Nice
Jan. 2017, Journées Niçoises de Mécanique des Fluides
Aug. 2015 , congrès général SFP, Strasbourg
Jun. 2015, GDR Mephy, Agay
Nov. 2012, GDR Liquides aux Interfaces, Lyon
Sep. 2012, GDR Mousse, Paris
Jun. 2012, GDR Mephy, Agay
Jun. 2011, Capillary day, Nice
Feb. 2010, Journées Niçoises de Mécanique des Fluides
Aug. 2009, congrès français de mécanique, Marseille
Aug. 2007 , congrès français de mécanique, Grenoble

SELECTED ABSTRACTS AT CONFERENCES
[120] Feb. 2025, Complex Days, Nice

Dynamic meniscus in contact with a vertical soap film
A. Vigna-Brummer*, A. Monier, Isabelle Cantat, C. Brouzet and C. Raufaste
[119] Oct. 2024, JMC 2024, Marseille
Studies on the circular hydraulic jump
K. Ongari*, C. Raufaste, F. Celestini and M. Argentina
[118] Oct. 2024, JMC 2024, Marseille
Isolated meniscus around rings suspended in vertical soap films
A. Vigna-Brummer*, A. Monier, C. Brouzet and C. Raufaste
[117] Oct. 2024, JMC 2024, Marseille
Mobile soap film spinning
A. Monier*, K. Gutierrez, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste
[116] Oct. 2024, GFR 2024, Marseille
Stress redistribution in flowing liquid foam
F. Schott*, S. Gstöhl, C. M. Schlepütz, B. Dollet, C. Raufaste, S. Santucci, C. Claudet and R. Mokso
[115] Oct. 2024, INES 2024, Nice
Shape and dynamics of menisci at the contact with soap films
A. Monier, F.-X. Gauci, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste*
[114] Sep. 2024, EFDC1 2024, Aachen
Drainage of mobile soap films under gravity: characterizing the space-time dynamics
C. Raufaste*, A. Monier, C. Claudet, F. Celestini and C. Brouzet
[113] Sep. 2024, EFDC1 2024, Aachen
Optimum control strategies for maximum thrust production in underwater undulatory swimming
M. Argentina*, L. Fu, S. Israilov, J. Sanchéz Rodriguéz, C. Brouzet, G. Allibert and C. Raufaste
[112] Sep. 2024, EFDC1 2024, Aachen
Local stress measurements in the elastoplastic regime of a flowing sheared foam
B. Dollet*, F. Schott, S. Gstöhl, C. Schlepütz, C. Raufaste, S. Santucci, C. Claudet and R. Mokso
[111] Sep. 2024, 38th Conference of the European Colloid & Interface Society (ECIS2024), Copenhagen
Stress redistribution in flowing liquid foam
F. Schott*, S. Gstöhl, C. Schlepütz, B. Dollet, C. Raufaste, C. Claudet, S. Santucci and R. Mokso
[110] July 2024, Eufoam 2024, Dresden
Soap film interface mobility sets spinning drainage dynamics
A. Monier*, K. Gutierrez, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste
[109] July 2024, Eufoam 2024, Dresden
Stress redistribution in flowing liquid foam
F. Schott*, S. Gstöhl, C. M. Schlepütz, B. Dollet, C. Raufaste, S. Santucci, C. Claudet and R. Mokso
[108] July 2024, Eufoam 2024, Dresden
Meniscus size around rings suspended in vertical soap films
A. Vigna-Brummer*, A. Monier, C. Brouzet and C. Raufaste
[107] July 2024, Eufoam 2024, Dresden

Space-time dynamics in gravity drainage of mobile soap films
A. Monier, F.-X. Gauci, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste*
[106] July 2024, Eufoam 2024, Dresden
Imaging 3D foam relaxation after shear cessation
F. Schott, S. Gstöhl, C. M. Schlepütz, C. Raufaste, S. Santucci, C. Claudet, R. Mokso and B. Dollet*
[105] June 2024, Soft Matter meeting 2024, Strasbourg
Drainage of vertical soap films
A. Monier, F.-X. Gauci, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste*
[104] June 2024, Fluid and Elasticity 2024, Arcachon
Optimum control strategies for maximum thrust production in underwater undulatory swimming
L. Fu, S. Israilov, J. Sanchez-Rodriguez, C. Brouzet, G. Allibert, C Raufaste and M. Argentina*
[103] Dec. 2023, Fluids and Complexity, Nice
Gravitational study on mobile soap film drainage in solid frames
A. Monier*, F.-X. Gauci, K. Gutierrez, C. Claudet, C. Brouzet, F. Celestini and C. Raufaste
[102] Dec. 2023, Fluids and Complexity, Nice
Optimum control strategies for maximum thrust production in underwater undulatory swimming
L. Fu*, S. Israilov, J. Sanchez-Rodriguez, C. Brouzet, G. Allibert, C Raufaste and M. Argentina
[101] Dec. 2023, Fluids and Complexity, Nice
Wake structure of a robotic fish
C. Brouzet*, C. Raufaste and M. Argentina
[100] Dec. 2023, Fluids and Complexity, Nice
Bouncing of Leidenfrost jets
D. Paulovics*, C. Raufaste, T. Frisch and F. Celestini
[99] Nov. 2023, APS DFD 2023, Washington DC
Impact and bouncing of Leidenfrost Jets
D. Paulovics, C. Raufaste, T. Frisch and F. Celestini*
[98] Nov. 2023, APS DFD 2023, Washington DC
Frost propagation on breath figures
D. Paulovics*, C. Raufaste, T. Frisch, C. Claudel and F. Celestini
[97] Jul. 2023, Internation Congress on Rheology, Athens
Simultaneous micro- and macroscopic description of foam shear in a plate-plate rheometer with X-ray microscopic tomography
F. Schott*, S.J. Gstöhl, C.M. Schlepütz, B. Dollet , C.Raufaste , S. Santucci, C. Claudet and R. Mokso
[96] Jul. 2023, Congrès annuel de la SFP, Paris
Throwing soft objects with the superpropulsion effect
G. Giombini, C. D'Angelo, F. Celestini and C. Raufaste*
[95] June 2023, 11th Conference of the International Marangoni Association, Bordeaux, France
Characterizing the gravity drainage of mobile soap films
A. Monier*, F.-X. Gauci, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste
[94] June 2023, Workshop Mousses, films et effets Marangoni, Rennes
Drainage de films de savon aux interfaces mobiles, avec ménisque II
A. Monier, F.-X. Gauci, C. Brouzet, C. Claudet, F. Celestini and C. Raufaste*

[93] June 2023, Workshop Mousses, films et effets Marangoni, Rennes
Drainage de films de savon aux interfaces mobiles, avec ménisque I
A. Monier*, F.-X. Gauci, C. Brouzet, C. Claudet, F. Celestini and C. Raufaste
[92] June 2023, 11th Conference of the International Marangoni Association, Bordeaux, France
Characterizing the gravity drainage of mobile soap films
A. Monier*, F.-X. Gauci, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste
[91] Mar. 2023, Rencontres du non linéaire, Paris
Gravitational drainage of mobile soap films
A. Monier*, F.-X. Gauci, C. Claudet, F. Celestini, C. Brouzet and C. Raufaste
[90] Mar. 2023, 3S'23, Symposium on Surface Science, Courmayeur
Atomistic description of interdroplet ice-bridge formation during condensation frosting
S. Curiotto, D. Paulovics, C. Raufaste, F. Celestini, T. Frisch, F. Leroy, F. Cheynis and P. Müller*
[89] Nov. 2022, 1ères journées du GDR IDE "Interaction Désordre Elasticité"
Imaging three-dimensional (3D) foam flows in real time at the bubble scale using fast X-ray tomography
F. Schott, B. Dollet*, S. Santucci, C. Claudet, M. Argentina, C. Raufaste, R. Mokso
[88] Nov. 2022, GDR Navier-Stokes 2.00
Wake structure of a robotic fish
C. Brouzet*, C. Raufaste, T. Frisch, M. Argentina
[87] Oct. 2022, 56ème congrès du Groupe Français de Rhéologie
3D liquid foam flow through an hourglass-like constriction
F. Schott*, C. Raufaste, B. Dollet, S. Santucci, C. Claudet, S. Gstöhl, R. Mokso
[86] Oct. 2022, 56ème congrès du Groupe Français de Rhéologie
Drainage d'un film de savon en rotation
A. Monier*, C. Brouzet, F. Celestini, C. Claudet, C. Raufaste
[85] Sep. 2022, EFMC14 (14th European Fluid Mechanics Conference), Athens
Tail beat frequency selection for inertial underwater swimmers
J. Sanchez-Rodriguez*, C. Raufaste, M. Argentina
[84] July 2022, Eufoam 2022, Krakow
3D liquid foam flow in an hourglass-like constriction
F. Schott*, C. Raufaste, B. Dollet, S. Santucci, C. Claudet, S.J. Gstöhl and Rajmund Mokso
[83] July 2022, Eufoam 2022, Krakow
Why is X-ray tomographic microscopy a useful tool in foam characterization?
R. Mokso, F. Schott, C. Raufaste, B. Dollet, S. Santucci
[82] July 2022, Eufoam 2022, Krakow
Prediction of confined foam flows in complex geometries: the example of the negative wake
B. Dollet, L. Rose, C. Raufaste and S. Santucci
[81] July 2022, Eufoam 2022, Krakow
Coupling the shape and motion of bubbles in a confined 2D flowing foam
B. Dollet, L. Rose, S. Santucci and C. Raufaste*
[80] June 2022, 5th International Conference on Tomography of Materials & Structures, Grenoble
FOAMQUANT: a 4D quantification toolbox
F. Schott*, C. Raufaste, B. Dollet, S. Santucci, C. Claudet and R. Mokso

[79] June 2022, 5th International Conference on Tomography of Materials & Structures, Grenoble
Tomo-rheoscopy of a 3D liquid foam
F. Schott*, C. Raufaste, B. Dollet, S. Santucci, C. Claudet and R. Mokso
[78] May 2022, Foam-Scatter workshop, Grenoble
Imaging 3D foam flows using fast X-ray tomography: hopper flow and plate-plate rheometer
F. Schott, C. Raufaste, B. Dollet*, S. Santucci, C. Claudet and R. Mokso
[77] May 2022, Journées givrage-mouillage du GDR TRANSINTER
Frost propagation on breath figures
D. Paulovics, C. Raufaste, T. Frisch, C. Claudet, F. Celestini
[76] Mar 2022, Rencontres du non linéaire
Frost propagation on breath figures
D. Paulovics, C. Raufaste, T. Frisch, C. Claudet, F. Celestini
[75] Mar 2022, Workshop systèmes à 3 phases du GDR MEPHY
Frost propagation on breath figures
D. Paulovics, C. Raufaste, T. Frisch, C. Claudet, F. Celestini
[74] Nov. 2021, Fluids and Complexity II, Nice
Swimming gait optimized by Machine Learning
L. Fu, J. Sanchez-Rodriguez, S. Israilov, G. Allibert, C. Raufaste and M. Argentina
[73] Nov. 2021, Fluids and Complexity II, Nice
Characterization of bubbles deformations and interactions inside a liquid foam flowing through a constriction
F. Schott, C. Raufaste, B. Dollet, S. Santucci, C. Claudet and R. Mokso
[72] Nov. 2021, Fluids and Complexity II, Nice
Swimming gait driven by proprioception
J. Sanchez-Rodriguez, C. Raufaste and M. Argentina
[71] Aug. 2021, 25th Congress ICTAM Milano 2020+1 Virtual
A simple model for fish locomotion: from kinematics to proprioceptive driving.
J. Sánchez-Rodríguez, C. Raufaste, M. Argentina.
[70] Apr. 2021, AERC 2021, Annual European Rheology Conference in Cyberspace (virtual)
Elastic overshoot in liquid foam flows through and around a permeable defect
L. Rose, B. Dollet, C. Raufaste and S. Santucci
[69] Mar. 2021, Rencontre du Non Linéaire 2021
Superpropulsion of slender elastic objects
G. Giombini, F. Celestini and C. Raufaste
[68] Mar. 2021, APS March Meeting 2021 (virtual)
A proprioceptive mechanism for swimming
J. Sanchez Rodriguez, C. Raufaste and M. Argentina
[67] Mar. 2021, APS March Meeting 2021 (virtual)
Intermediate soft layer for optimizing throws
G. Giombini, J. Mathiesen, M. Argentina, C. D'Angelo, F. Celestini and C. Raufaste
[66] Mar. 2021, APS March Meeting 2021 (virtual)
Impacts of bi-layered elastic projectiles
C. D'Angelo, F. Celestini and C. Raufaste
[65] Jul. 2020, Eufoam 2020, Aberystwyth (virtual)

Interaction between a descending solid object and a soap film
I.T. Davies and C. Raufaste

[64] Jul. 2020, Eufoam 2020, Aberystwyth (virtual)
Liquid foam flow through/around a permeable defect
L. Rose, N. Shmakova, C. Raufaste, B. Dollet, S. Santucci

[63] Jul. 2020, Eufoam 2020, Aberystwyth (virtual)
Geometrical evolution during the collapse of soap-film catenoids
C. Raufaste, A.I. Pesci, J.D. Shemilt, S.J. Cox, R.E. Goldstein

[62] Nov 2019, APS DFD 2019, Seattle
Swimming gait driven by proprioceptive feedback
J. Sanchez-Rodriguez, C. Raufaste, M. Argentina

[61] Nov. 2019, APS DFD 2019, Seattle,
A proprioceptive robotic swimmer
M. Argentina, J. Sanchez Rodriguez, C. Raufaste

[60] Nov. 2019, GDR SLAMM, Roscoff
Impacts of bicomposite projectiles
C. D'Angelo, F. Celestini and C.Raufaste

[59] Jun. 2019, Waves Côte d'Azur, Nice
Nonlinear waves in Plateau borders
C. Raufaste

[58] Mar. 2019, Complex Days, Nice
The complex dynamics of bouncing
C.D'Angelo, C.Raufaste and F.Celestini

[57] Mar. 2019, Complex Days, Nice
Bounces of bicomposite object
C. D'Angelo, C. Raufaste and F. Celestini

[56] Jan. 2019, Journée de Physique Statistique et de la Matière Molle, Paris
Impacts of bicomposite projectiles
C. D'Angelo, C. Raufaste and F. Celestini

[55] Dec. 2018, Fluids and complexity, Nice
The superpropulsion effect with droplets and soft elastic projectiles
C. Raufaste, G. Ramos Chagas, T. Darmanin, C. Claudet, F. Guittard, F. Celestini

[54] Nov. 2018, APS DFD 2018, Atlanta, US
Study of the thrust-drag balance with a swimming robotic fish
M. Argentina, F. Gibouin, C. Raufaste

[53] Jul. 2018, Eufoam 2018, Liège
Stability and relaxation dynamics of a single Plateau border
C. Raufaste

[52] Dec. 2017, Nice Nonlineartities, Nice
Flow and nonlinearities inside the liquid foam microchannels
A. Cohen, N. Fraysse, J. Rajchenbach, Y. Bouret, M. Argentina, C. Raufaste

[51] Nov. 2017, APS DFD 2017, Denver, US
Scaling laws for a compliant biomimetic swimmer
F. Gibouin, C. Raufaste, Y. Bouret, M. Argentina

[50] Nov. 2017, APS DFD 2017, Denver, US
Hydraulic jumps in the liquid foam microchannels
C. Raufaste, A. Cohen, N. Fraysse, J. Rajchenbach, Y. Bouret, M. Argentina
[49] Jan. 2017, Journées Niçoises de Mécanique des Fluides, Nice
Inertial dynamics of a liquid string
A. Cohen, N. Fraysse, M. Argentina, Y. Bouret, C. Raufaste
[48] June 2016, Statistical physics of materials, Aussois
Foams imbibition in a fracture
T. Chevalier, M. J. Alava, A. Puisto, C. Raufaste, S. Santucci
[47] August 2015, congrès annuel SFP, Strasbourg
Two dimensional Leidenfrost droplets in a Hele-Shaw cell
F. Celestini, A. Cohen, A. Barzyk, C. Raufaste, T. Frisch, L. Duchemin, Y. Pomeau
[46] June 2015, GDR Mephy, Agay
Inertial flow in a Plateau Border
A. Cohen, N. Fraysse, J. Rajchenbach, M. Argentina, Y. Bouret, C. Raufaste
[45] June 2015, GDR Mephy, Agay
2D Leidenfrost droplet in a Hele-Shaw cell
F. Celestini, A. Cohen, A. Barzyk, C. Raufaste, T. Frisch, L. Duchemin, Y. Pomeau
[44] March 2015, FloMat (Flowing Matter across the scales), Rome, Italy
Inertial flows in liquid foam microchannels
A. Cohen, N. Fraysse, J. Rajchenbach, M. Argentina, Y. Bouret, C. Raufaste
[43] August 2014, Condensed Matter in Paris 2014, JMC 14 - CMD 25
Capillary-hydraulic jump in liquid foams
A. Cohen, N. Fraysse, J. Rajchenbach, M. Argentina, Y. Bouret, C. Raufaste
[42] July 2014, Eufoam 2014, Thessaloniki, Greece
Capillary hydraulic jump in a Plateau border
A. Cohen, N. Fraysse, J. Rajchenbach, M. Argentina, Y. Bouret, C. Raufaste
[41] July 2014, Eufoam 2014, Thessaloniki, Greece
Four dimensional foam studies at the micrometer scale
R. Mokso, K. Mader, B. Dollet, S. Santucci, G. Lovric, C. Raufaste
[40] June 2013, GDR Mousse, Paris
Bord de Plateau en régime harmonique
A. Cohen, N. Fraysse, J. Rajchenbach, C. Raufaste
[39] May 2013, workshop on Patterns and hydrodynamic instabilities in reactive systems, Brussels, Belgium
Highly ordered pattern formation during chemical alterations
C. Raufaste, B. Jamtveit, T. John, P. Meakin, D. K. Dysthe
[38] November 2012, GDR Liquides aux Interfaces, Lyon
Imbibition inside a flexible pore
A. Cohen, N. Fraysse, J. Rajchenbach, C. Raufaste
[37] September 2012, GDR Mousse, Paris
Fast dynamics in liquid foams
A. Cohen, G. Kirstetter, F. Celestini, N. Fraysse, J. Rajchenbach, C. Raufaste
[36] August 2012, Journée de la Matière Condensée, Montpellier

Bubble dynamics in complex fluids
F. Haudin, C. Raufaste, X. Noblin

[35] July 2012, Eufoam 2012, Lisbon, Portugal
Jet refraction across a soap film
C. Raufaste, G. Kirstetter, F. Celestini, S. Cox

[34] July 2012, Eufoam 2012, Lisbon, Portugal
Drop coalescence on a single Plateau border
A. Cohen, N. Fraysse, J. Rajchenbach, C. Raufaste

[33] July 2012, Eufoam 2012, Lisbon, Portugal
Stretching the limits of 4D observation of foam dynamics
R. Mokso, K. Mader, B. Dollet, S. Santucci, C. Raufaste

[32] July 2012, Eufoam 2012, Lisbon, Portugal
Linking structure and flow inside 3D foams
S. Santucci, K. Mader, R. Mokso, B. Dollet, C. Raufaste

[31] July 2012, Eufoam 2012, Lisbon, Portugal
Automated, high throughput structural characterization of liquid foams
K. Mader, R. Mokso, C. Raufaste, B. Dollet, S. Santucci, M. Stampanoni

[30] June 2012, GDR Mephy, Agay
Morphologies and mechanisms during chemical alterations
C. Raufaste, B. Jamtveit, T. John, P. Meakin, D. K. Dysthe

[29] June 2012, GDR Mephy, Agay
Coalescence de gouttes dans un bord de Plateau
A. Cohen, N. Fraysse, J. Rajchenbach, C. Raufaste

[28] Jan. 2012, CECAM workshop, dissipative rheology of foams
Drop coalescence along a Plateau Border
A. Cohen, N. Fraysse, C. Raufaste

[27] Jan. 2012, CECAM workshop, dissipative rheology of foams
Impact of a liquid jet on a soap film
G. Kirstetter, C. Raufaste, F. Celestini

[26] Sep. 2011, Jump 2011, PSI user meeting, Villigen, Switzerland
Imaging and mechanics of liquid foam flows
K. Mader, R. Mokso, B. Dollet, S. Santucci, C. Raufaste

[25] Sep. 2011, Liquid Matter, Wien, Austria
Interaction of a liquid jet with a soap film
G. Kirstetter, C. Raufaste, F. Celestini

[24] Sep. 2011, Liquid Matter, Wien, Austria
Bubble dynamics in complex fluids
F. Haudin, C. Raufaste, X. Noblin

[23] Sep. 2011, Liquid Matter, Wien, Austria
Role of the fluid and porosity formation during solvent-mediated phase transformations
C. Raufaste, B. Jamtveit, T. John, P. Meakin, D.K. Dysthe

[22] July 2011, SFP, Bordeaux, France
Impact d'un jet sur un film liquide
G. Kirstetter, C. Raufaste, F. Celestini, J. Rajchenbach

[21] June 2011, Capillary day, Nice, France
Interaction of a liquid jet with a soap film
G. Kirstetter, C. Raufaste, F. Celestini
[20] May 2011, IUTAM Symposium: Mechanics of liquid and solid foams, Austin, US
Interaction of a liquid jet with a liquid foam : a film scale study
C. Raufaste, G. Kirstetter, F. Celestini
[19] Jan. 2011, 31èmes Journées de Physique Statistique, Paris, France
Impact d'un jet sur un film liquide
G. Kirstetter, F. Celestini, C. Raufaste, J. Rajchenbach
[18] Jan. 2011, Geological carbon capture & storage in mafic and ultramaic rocks, IODP/ICDP Workshop on the role of oceanic and continental scientific drilling, Muscat, Oman
Physical feedbacks during mineral carbonation: Field observations, experiments, and models
B. Jamtveit, A. Malthe-Sørenssen, H. Austrheim, A. Røyne, C. Raufaste
[17] Jul. 2010, Eufoam 2010, Borovets, Bulgaria
Orientational effects in the flow of liquid foams
C. Raufaste, S.J. Cox, P. Marmottant, F. Graner
[16] June 2010, Cryspom II, Brienz, Switzerland
Crystal growth and porosity formation during solvent-mediated phase transformations
C. Raufaste, B. Jamtveit, T. John, D. K. Dysthe
[15] June 2010, Cryspom II, Brienz, Switzerland
Salt crystallization induced by evaporation in porous media
M. Angeli, D. K. Dysthe, C. Raufaste
[14] Jan. 2010, 29th Nordic Geological Winter Meeting, Oslo, Norway
Mechanisms of permeability enhancement during chemical alteration
C. Raufaste, J. Mathiesen, B. Jamtveit, T. John, D. K. Dysthe
[13] Jan. 2010, 29th Nordic Geological Winter Meeting, Oslo, Norway
The Andean Geotrail (1): A scientific adventure
S. Caroline, G. Olivier, C. Raufaste, K. Mair
[12] Dec. 2009, AGU fall meeting 2009, San Francisco, US
The Andean Geotrail (1): A scientific adventure
S. Caroline, G. Olivier, C. Raufaste, K. Mair
[11] Aug. 2009, 19th french congress of mechanics, Marseille, France
Volume changes in solids induced by chemical alteration
C. Raufaste, J. Mathiesen, A. Røyne, D. K. Dysthe, A. Malthe-Sørenssen, B. Jamtveit
[10] Aug. 2009, 19th french congress of mechanics, Marseille, France
Evaporation et cristallisation de sels solubles dans un réseau poreux modèle
M. Angeli, C. Raufaste, D. K. Dysthe
[9] May 2009, the 22nd Kongsberg seminar, Kongsberg, Norway
Pattern formation in a fluidized bimodal granular mixture
A. Nermoen, D. K. Dysthe, E. Jettestuen, C. Raufaste
[8] Mar. 2009, Geilo School, Geilo, Norway
Coupling between chemical alteration and mechanics: an experimental approach
C. Raufaste
[7] Mar. 2009, Interpore conference, Kaiserslautern, Germany 2009

Sodium sulphate crystallization experiment in sintered glass
M. Angeli, C. Raufaste, D. K. Dysthe

[6] Oct. 2008, 43ème Colloque du Groupe Français de Rhéologie, Palaiseau, France
Rhéologie et imagerie des écoulements 2D de mousse : de la bulle au milieu continu
C. Raufaste, I. Cheddadi, P. Marmottant, P. Saramito and F. Graner

[5] July 2008, Eufoam 2008, Noordwijk, The Netherlands
Numerical modelling of foam Couette flows
I.Cheddadi, P. Saramito, C. Raufaste, P. Marmottant and F. Graner

[4] Oct. 2007, Visco-plastic fluids: from theory to application, Monte Verità, Tessin, Switzerland
Couette flow of a foam: numerical simulation and experiments
I. Cheddadi, P. Saramito, F. Graner, P. Marmottant and C. Raufaste

[3] Aug. 2007, 18th french congress of mechanics, Grenoble, France
Ecoulement 2D de mousse : une description locale
C. Raufaste, P. Marmottant, F. Graner

[2] Aug. 2007, 18th french congress of mechanics, Grenoble, France
Interface dynamique entre liquides miscibles
P. Philippe, C. Raufaste, P. Kurowski, P. Petitjeans

[1] July 2006, Eufoam 2006, Potsdam, Germany
The complex behaviour of 2D foam flow: a local description
C. Raufaste, P. Marmottant, F. Graner

TEACHING, LECTURES

Responsibilities

Responsible for teaching modules
since 2018: Fluid mechanics (Master 1 OAM), 6 ECTS credits, management of a team of 3 teachers and of approx. 10 students
2015-2018: Applied thermodynamics (Master 1 Pro EME), 3 ECTS credits, approx. 20 students
2014-2018: Mathematics (Master 1 Pro EME), 3 ECTS credits, approx. 20 students

Responsible for lab sessions
2015-2020: Foam physics (M2 P3M), design and implementation of 2 experiments, management of a team of 2 teachers, approx. 10 students
2014-2019: Waves (L2), design and implementation of 3 experiments, management of a team of 3 teachers, approx. 50 students

Administrative responsibilities
since 2018: member of the steering committee of the Master OAM ("Ondes, Atomes et Matière", Université Côte d'Azur): creation of the Master; choice of the program, teaching modules and pedagogical teams; student and teacher management. http://master-oam.unice.fr/

Master's degree

Fluid mechanics (M1 OAM, since 2018)

Lectures (12 hours per year)
Exercises (16 hours per year)
approx. 10 students per year.

* Responsible for the teaching module (6 ECTS credits)
* Management of a team of 3 teachers and of approx. 10 students.
* Design of the module from scratch
* Courses on the basics of fluid mechanics, viscous flows, potential flows and surface waves.
* Design/correction of evaluations

Foam physics (M2 P3M, 2015-2020)

Lectures (3 hours per year)
Lab sessions (12 hours per year)
approx. 10 students

* Responsible for the teaching module (3 ECTS credits) and for the lab sessions.
* Management of a team of 2 teachers and of approx. 10 students.
* Design of the module from scratch
* Introduction to the physics of liquid foams (composition, Plateau's laws, mechanical properties)

* Design and implementation of 2 experiments: Stokes experiment in a liquid foam (equivalent to the classical Stokes experiment of an object sedimenting in a Newtonian fluid, but here the fluid is non Newtonian) and bubble growth in a supersaturated liquid leading to foam formation.
* Writing of a lab manual.
* Supervision of students and correction of lab reports.

Statistical physics (M1 P3M, 2018/2019)

Lectures (15 hours per year)
Exercises (12 hours per year)
Lab sessions (20 hours per year)
approx. 10 students.

* Responsible for the teaching module (6 ECTS credits).
* Management of a team of 2 teachers and of approx. 10 students.
* Design of the lecture from scratch.
* Design/correction of evaluations.

Applied thermodynamics (M1 Pro EME, 2015-2018)

Lectures (7.5 hours per year)
Exercises (7.5 hours per year)
approx. 20 students

* Responsible for the teaching module
* Design of the lecture from scratch
* Design/correction of evaluations

Mathematics (M1 Pro EME, 2012-2018)

Lectures (9 hours per year)
Exercises (9 hours per year)
approx. 20 students

* Responsible for the teaching module
* Module designed for students having difficulties in mathematics to bring them to the required level of this Master's degree.
* Writing of a lecture document
* Mixed approach: classical lecture and project-based sessions. I divided the students into groups of 3-4 students, each of them with a rather open subject (Gini coefficient of a company, Cobb-Douglas production function, ...). They had to find the proper mathematical tools to solve the problem and to defend their project.
* Design/correction of evaluations

Soft matter (M1 P3M, 2009-2017)

Lectures (12 hours per year)
Exercises (6 hours per year)
Lab sessions (4 hours per year)
approx. 8 students per year

* Teaching module about fluid interfaces, surface tension and soft rheology

* Design of the lecture from scratch
* Design and supervision of 1 lab session (Stokes flow in microchannels studied by microscopy and by particle image velocimetry)
* Design/correction of evaluations

Electricity (M1 EFTIS prepa CAPES/AGREG, 2010-2012)

Lectures/Exercises (18 hours per year), approx. 10 students per year
* Responsible for the teaching module
* Module designed from scratch
* Design/correction of evaluations

Fluid mechanics (M2 EFTIS prepa CAPES/AGREG, 2009-2012)

Online lectures (40 hours per year), approx. 10 students per year
* Responsible for the teaching module
* Module designed from scratch
* Writing of a lecture document
* Online lectures ("Master à distance") on fluid mechanics to prepare the teaching certification
* Design/correction of evaluations

Bachelor's degree

Lab project (L3, since 2013)

Equivalent to 15 teaching hours per year
* Supervision of 2-3 students to perform research-like experiments in a research institute. The project lasts 1 month approximately.
* Development of a new experiment every year.
* Examples

De bien singuliers rebonds
http://physique.unice.fr/sem6/2019-2020/PagesWeb/PT/Rebond.pdf
La Physique du propulseur: une arme préhistorique
http://physique.unice.fr/sem6/2017-2018/PagesWeb/PT/Propulseur/
Catapulte intelligente
http://physique.unice.fr/sem6/2016-2017/PagesWeb/PT/Catapulte/
Gouttes de Leidenfrost paramagnétiques
http://physique.unice.fr/sem6/2015-2016/PagesWeb/PT/Leidenfrost/
Acoustique des mousses liquides
http://physique.unice.fr/sem6/2013-2014/PagesWeb/PT/Mousses/

Physics of waves (L2, 2014-2019)

Lab sessions (36 hours per year)
* Responsible of the lab sessions. Management of a team of 3 teachers, approx. 50 students per year.

* New module initially, everything was designed from scratch
* Design and implementation of 3 experiments: wave propagation (ultrasonic transducers ; travel time, wavelength and amplitude attenuation measurements ; application to echography), wave diffraction (ultrasonic transducer, characterization of the emission cone, two waves interferences) and wave interferences (Melde's and Kundt's tube experiments).
* Writing of a lab manual.

Continuum mechanics (L3, 2009-2020)

Lab sessions (36 hours per year), approx. 30 students per year

* Supervision of students and correction of lab reports.

Physics for biologists (L1, 2012-2015)

Lectures/Exercises (24 hours per year), approx. 25 students per year

* Teaching the basics of fluid mechanics and thermodynamics for students in biology.
* Design/correction of evaluations

Experimental physics (L2, 2009-2011)

Lab sessions (27 hours per year), approx 25 students per year

* Supervision of students and correction of lab reports.